



DUNMAN SECONDARY SCHOOL

CANDIDATE
NAME

CLASS

INDEX
NUMBER

PRELIMINARY EXAMINATION 2024 SECONDARY 4 EXPRESS / 5 NORMAL ACADEMIC

ADDITIONAL MATHEMATICS

Paper 1

4049/01

20 August 2024
2 hours 15 minutes

Marking Scheme

(Updated 1 Dec 2022 by HOD/Math aligned to Marking Scheme for Specimen Papers 4049)

Question	Answer	Marks	Partial Marks	Guidance	Marker's Comments/ Reminders to Students
1	$h = \frac{7+3\sqrt{5}}{4+2\sqrt{5}} \times \frac{4-2\sqrt{5}}{4-2\sqrt{5}}$	3	M1	Correct rationalisation	Students' Performance (P): Well done, showing a strong understanding of rationalization.
	$= \frac{28-14\sqrt{5}+12\sqrt{5}-30}{-4}$		M1✓	Their expansion of numerator	
	$= \frac{-2-2\sqrt{5}}{-4}$ $= \frac{1}{2} + \frac{1}{2}\sqrt{5}$		A1	C.A.O	

Question	Answer	Marks	Partial Marks	Guidance	
2	$y = -4x^2 + 12x - 3$ $= -4(x^2 - 3x) - 3$ $= -4\left[\left(x - 1\frac{1}{2}\right)^2 - \left(1\frac{1}{2}\right)^2\right] - 3$	4	M1	Correct completion of square	Students' Performance (P): Average performance, with many students losing marks due to completing the square Common mistakes (CM): - Remembering to factorise out -4 but forgot to include it back, resulting in $-4\left(x - 1\frac{1}{2}\right)^2 - 5\frac{1}{2}$ or $-4\left(x - 1\frac{1}{2}\right)^2 - \frac{1}{2}$ "$-4x^2 + 12x - 3$ is always less than or equal to 6" rather than saying "y is always less than or equal to 6", though not penalized this time - Trying to 'force' the explanation for the 2nd part of the question when their completed square form is wrong Reminders for students (R): Students need to understand that when it comes to explanation questions, they need to be clear.
	$= -4\left(x - 1\frac{1}{2}\right)^2 + 9 - 3$ $= -4\left(x - 1\frac{1}{2}\right)^2 + 6$		A1	Correct form OR B2, -1 mark for each error	
	$\left(x - 1\frac{1}{2}\right)^2 \geq 0$ $-4\left(x - 1\frac{1}{2}\right)^2 \leq 0$		M1	$\left(x - 1\frac{1}{2}\right)^2 \geq 0$ seen or better	
	$-4\left(x - 1\frac{1}{2}\right)^2 + 6 \leq 6$ $y \leq 6$		A1		
	Alternative Method Since coefficient of $x^2 < 0$, it means that the curve opens downwards		B1		
	and maximum value of y is 6, $y \leq 6$		B1	state the maximum value of y is 6	

Question	Answer	Marks	Partial Marks	Guidance	
3(a)	$f(-1) = R$ $-a + b + 5 + 7 = R$ $-a + b + 12 = R$ $f(2) = 2R - 1$ $8a + 4b - 10 + 7 = 2R - 1$ $8a + 4b - 3 = 2R - 1$	3	M1	Either substitution correct using Remainder Theorem	Students' Performance (P): Very well done, showing most candidates understand the application of Remainder Theorem. Only a slight handful of students did Long Division; while it is a valid method, it is recommended not to do so as it can be tedious, resulting in algebraic slips.
	$8a + 4b - 3 = 2(-a + b + 12) - 1$ $8a + 4b - 3 = -2a + 2b + 23$ $10a + 2b = 26$		M1	Either substitution or elimination method seen	
	$5a + b = 13$		A1		

Question	Answer	Marks	Partial Marks	Guidance	
3(b)	$ab = 6$ $a = \frac{6}{b}$ $5\left(\frac{6}{b}\right) + b = 13$	3	M1	Sub either equation into the other	Students' Performance (P): Well done, with some students using the factors of 6 to attempt Guess and Check Common mistakes (CM): Did not reject either $a = 0.6$ and just conclude only 1 pair of answer
	$30 + b^2 = 13b$ $b^2 - 13b + 30 = 0$ $(b - 10)(b - 3) = 0$		M1✓	Quadratic Factorisation / Formula applied into equation	
	$b = 3$ or 10 (rej, as $b > a$) $a = 2$		A1	Correct pair of answer (inclusive of rejecting 10)	
	Alternative Method $a(13 - 5a) = 6$ $5a^2 - 13a + 6 = 0$ $(5a - 3)(a - 2) = 0$ $a = \frac{3}{5}$ (rej, as a must be an integer) or 2 $b = 10$ or 3		M1 M1✓ A1	Sub (a)'s equation into (b) Factorise correctly Correct pair of answer (inclusive of rejecting $\frac{3}{5}$)	
	Alternative Method M1 – Breaking down factors of ab M1 – guess & check (with verification seen) A1				

Question	Answer	Marks	Partial Marks	Guidance	
4	<u>If workings sighted:</u> $\int \frac{4}{x^3} + \frac{3}{3x-7} dx$ $= 4 \left(\frac{x^{-2}}{-2} \right) + 3 \left(\frac{\ln(3x-7)}{3} \right) + c$ $= -\frac{2}{x^2} + \ln(3x-7) + c$	4	M1	x^{-2} seen	<u>Students' Performance (P):</u> Well done, showing most students can do indefinite integrals <u>Common mistakes (CM):</u> • Forgot that it is divided by 3, and instead multiplied by 3, ending up with $9\ln(3x-7)$ • Forgot +c in their workings
			M1	$\ln(3x-7)$ seen	
			A1	$-\frac{2}{x^2}$ seen (do not accept $-2x^{-2}$)	
			A1	$\ln(3x-7)$ seen (aka coefficient 1) -1 if +c is not seen throughout the integration *If go straight to $\ln(3x-7)$, Award B2	
	<u>If no workings sighted:</u> B1 for - 2 B1 for $\frac{1}{x^2}$ B2 for $\ln(3x-7)$				

Question	Answer	Marks	Partial Marks	Guidance	
5(a)	$\text{LHS} = 16 \sin \frac{x}{2} \cos \frac{x}{2} - 8 \cos 2x + 9$ $= 8 \left(2 \sin \frac{x}{2} \cos \frac{x}{2} \right) - 8(1 - 2 \sin^2 x) + 9$ $= 8(\sin x) - 8 + 16 \sin^2 x + 9$	4	M1	Convert $2 \sin \frac{x}{2} \cos \frac{x}{2}$ to $\sin x$ or better	Students' Performance (P): Average performance, with many students losing marks due to them not converting the double angle formula in the formula sheet into the half angle formula Reminders for students (R): Students need to realise that the double angle formula can be used not just for angles that are multiples of 2, e.g. $2x$, $8x$ etc, but it can also be used in multiples of $\frac{1}{2}$ e.g. $\frac{1}{2}x$, $\frac{1}{8}x$
	$= 16 \sin^2 x + 8 \sin x + 1$		M1	Convert $\cos 2x$ to $1 - 2 \sin^2 x$	
	$= (4 \sin x + 1)^2$		M1	Simplifying to $16 \sin^2 x + 8 \sin x + 1$	
	$= \text{RHS}$		A1	Usage of algebraic identity “LHS = = RHS” must be sighted for the final A1	
	Alternative Method $\text{RHS} = (4 \sin x + 1)^2$ $= 16 \sin^2 x + 8 \sin x + 1$ $= 8(1 - \cos 2x) + 8(2 \sin \frac{x}{2} \cos \frac{x}{2}) + 1$ $= 16 \sin \frac{x}{2} \cos \frac{x}{2} - 8 \cos 2x + 9$ $= \text{LHS}$	4	M1 M1 M1 A1	Expansion via algebraic identity Convert $\sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2}$ Convert $2 \sin^2 x = 1 - \cos 2x$ or better Simplification of expression “LHS = = RHS” must be sighted for the final A1	

Question	Answer	Marks	Partial Marks	Guidance	
5(b)	$16 \sin \frac{x}{2} \cos \frac{x}{2} - 8 \cos 2x + 9 = 9$ $(4 \sin x + 1)^2 = 9$ $4 \sin x + 1 = 3 \text{ or } -3$	3	M1	Obtaining $4 \sin x + 1 = 3 \text{ or } -3$	Students' Performance (P): Well done, with most candidates scoring full marks Common mistakes (CM): Having more than 3 answers, possibly due to the confusion with both answers having different signs, and thus thinking their angles should cover all 4 quadrants
	$\sin x = \frac{1}{2} \text{ or } \sin x = -1$ $\alpha = \frac{\pi}{6} \text{ or } \frac{\pi}{2}$				
	$x = \frac{\pi}{6}, \frac{5\pi}{6} \text{ or } x = \frac{3\pi}{2}$		A2	A1 for $x = \frac{\pi}{6}, \frac{5\pi}{6}$ A1 for $x = \frac{3\pi}{2}$ -1 if > 3 answers sighted, provided A2	
Question	Answer	Marks	Partial Marks	Guidance	
6	$\frac{4x^3 + 3x^2 + 2x + 1}{4x^3 - x^2} = 1 + \frac{4x^2 + 2x + 1}{4x^3 - x^2}$	6	M1	Obtaining 1 as quotient, and $4x^2 + 2x + 1$ as remainder after Long Division or better	Students' Performance (P): Poorly done, with a significant number of candidates scoring 3 Common mistakes (CM): - Did not attempt long division - Wrong partial fractions form Reminders for students (R): Students need to recall the definition of improper fractions as well as to recognize the various forms of partial fractions.
	$\frac{4x^2 + 2x + 1}{4x^3 - x^2} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{4x - 1}$		M1	Correct partial fractions form seen (provided LHS is proper)	
	$4x^2 + 2x + 1 = A(x)(4x - 1) + B(4x - 1) + C(x^2)$ Sub $x = 0, B = -1$		M1✓ B1✓	Express numerator as product of P.F. Correct value of B	
	Sub $x = \frac{1}{4}, C = 28$		B1✓	Correct value of C	
	Sub $x = 1, A = -6$		B1✓	Correct value of A	
	$\frac{4x^3 + 3x^2 + 2x + 1}{4x^3 - x^2} = 1 - \frac{6}{x} - \frac{1}{x^2} + \frac{28}{4x - 1}$			Minus 1 if final answer not written / copied wrongly	
	Max 1 out of 3 B1s if follow through				

Question	Answer	Marks	Partial Marks	Guidance	
7(a)	$R = \sqrt{3^2 + 4^2}$ $= 5$	4	M1	Correct substitution for R	Students' Performance (P): Poorly done, with a significant number of candidates scoring 2 Reminders for students (R): Students need to understand the question demands clearly, and not to attempt as if it is a routine question.
	$\alpha = \tan^{-1}\left(\frac{3}{4}\right)$ $= 36.869^\circ$		M1	Correct substitution for α	
	$3 \sin \theta + 4 \cos \theta = 5 \cos(\theta - 36.9^\circ)$		A1	5 seen	
			A1	36.9° seen (strictly 1 d.p.)	
Question	Answer	Marks	Partial Marks	Guidance	
7(b)	Smallest value is $50 - 25 = 25$ when $5 \cos(\theta - 36.9^\circ) = 5$	4	√B1	For correct smallest value	Students' Performance (P): Poorly done, despite follow-through marks awarded Common mistakes (CM): - Did not realise the question was asking for $(3 \sin \theta + 4 \cos \theta)^2$ - Did not realise the lower bound for $(3 \sin \theta + 4 \cos \theta)^2$ was 0
	$\cos(\theta - 36.9^\circ) = 1$ $\theta - 36.9^\circ = 0$ $\theta = 36.9^\circ$		√B1	For corresponding value of θ	
	Largest value is $50 - 0 = 50$ when $5 \cos(\theta - 36.9^\circ) = 0$		√B1	For correct largest value	
	$\cos(\theta - 36.9^\circ) = 0$ $\theta - 36.9^\circ = 90$ $\theta = 126.9^\circ$		√B1	For corresponding value of θ	

Question	Answer	Marks	Partial Marks	Guidance	
8(a)	$b = \frac{2\pi}{4\pi}$ $= \frac{1}{2}$	3	B1	Correct value of b	Students' Performance (P): Average performance
	$c = \frac{-7+1}{2}$ $= -3$		B1	Correct value of c	
	$1 = a \cos 0 - 3$ $a = 4$ $y = 4 \cos \frac{x}{2} - 3$		B1	Correct value of a -1 if final answer not in equation form (provided all 3 B1 obtained)	
Question	Answer	Marks	Partial Marks	Guidance	
8(b)	$a \cos bx + c = -1 - \frac{x}{\pi}$ $y = -1 - \frac{x}{\pi}$ Draw $y = -1 - \frac{x}{\pi}$	2	M1	Correct equation of line obtained	Students' Performance (P): Poorly done, with a significant number of candidates leaving the question blank Reminders for students (R): Students need to understand that EM concepts are pre-assumed knowledge in AM and can be tested without explicit labelling in the AM syllabus document.
	2 solutions		A1		

Question	Answer	Marks	Partial Marks	Guidance	
9(a)	Since $\frac{dh}{dt} = -\frac{t}{4}$, $h = \int -\frac{t}{4} dt = -\frac{t^2}{8} + c$	2	B1	Finding correct expression for h	Students' Performance (P): Well done with minor mistakes due to algebraic slips
	At $t = 0$, $h = 20$, $c = 20$.		B1	Correct value of c	
	$h = -\frac{t^2}{8} + 20$			-1 if did not express as $h = -\frac{t^2}{8} + 20$ (provided B2 obtained)	
Question	Answer	Marks	Partial Marks	Guidance	
9(b)	$\frac{dV}{dh} = \frac{3\pi h^2}{9}$	4	M1	Correct expression obtained	Students' Performance (P): Average performance, with most candidates scoring 2 out of 4 Common mistakes (CM): Forgot to include the power when replacing h with the expression from part (a) Reminders for students (R): Students need to be more careful when they do substitution
	At $t = 6$, $\frac{dh}{dt} = -\frac{6}{4} = -\frac{3}{2}$				
	$\frac{dV}{dt} = \frac{dV}{dh} \left(\frac{dh}{dt} \right)$ $= \frac{3\pi h^2}{9} \left(-\frac{3}{2} \right)$ $= \frac{3\pi}{9} \left(-\frac{t^2}{8} + 20 \right)^2 \left(-\frac{3}{2} \right) = -\frac{9\pi}{18} \left(-\frac{t^2}{8} + 20 \right)^2$		√M1 √M1	Chain Rule used with correct substitution from their $\frac{dV}{dh}$ Sub $t = 6$ into their $\frac{dV}{dt}$	
	When $t = 6$, $\frac{dV}{dt} = -377.38$ $= -377 \text{ cm}^3 / \text{s}$		A1	Accept $-120\frac{1}{8}\pi / -120.125\pi /$ $-\frac{961}{8}\pi$	

	<u>Alternative Method</u> $V = \frac{\pi}{9} \left(-\frac{t^2}{8} + 20 \right)^3$ $\frac{dV}{dt} = \frac{3\pi}{9} \left(-\frac{t^2}{8} + 20 \right)^2 \left(-\frac{2t}{8} \right)$ $= -\frac{\pi}{3} \left(-\frac{t^2}{8} + 20 \right)^2 \left(\frac{t}{4} \right)$ When $t = 6$, $\frac{dV}{dt} = -\frac{\pi}{3} \left(-\frac{6^2}{8} + 20 \right)^2 \left(\frac{6}{4} \right) = -377 \text{ cm}^3 / \text{s}$		M1 Forming expression of V in terms of t
		M1✓ Differentiate V with respect to t	
		M1✓ A1 Substitution of $t = 6$ Accept $-120\frac{1}{8}\pi / -120.125\pi$	

Question	Answer	Marks	Partial Marks	Guidance	
10(a)	$m_{AC} = \frac{4}{10}$ $= \frac{2}{5}$ $AC : y = \frac{2}{5}x + 10$	4	M1	Correct equation of line AC	Students' Performance (P): Did not do as well as expected, with a handful of candidates leaving the entire question blank Common mistakes (CM): Rejected the wrong x value from part (a)
	$\sqrt{145} = \sqrt{(x-4)^2 + \left(\frac{2}{5}x + 10\right)^2}$		M1✓	Equating given length to length formula using <i>their</i> equation of AC	
	$145 = x^2 - 8x + 16 + \frac{4}{25}x^2 + 8x + 100$ $3625 = 29x^2 + 2900$ $29x^2 - 725 = 0$ $x^2 - 25 = 0$ $x = \pm 5$		M1✓	Solving for x	
	C : (-5, 8)		A1	Both coordinates correct for A1	
Question	Answer	Marks	Partial Marks	Guidance	
10(b)	$\begin{array}{r rrrr} 1 & 10 & -5 & 4 & 10 \\ 2 & 14 & 8 & 0 & 14 \end{array}$	2	✓M1	Shoelace formula applied with workings shown	
	87 units ²		A1	C.A.O.	

Question	Answer	Marks	Partial Marks	Guidance	
11(a)	By Pythagoras' Theorem, $50^2 = (2x)^2 + (2y)^2$ $2500 = 4x^2 + 4y^2$	3	M1	Forming the correct equation based on PT	Students' Performance (P): Quite well done, with most candidates scoring full marks
	$y^2 = \frac{2500 - 4x^2}{4}$ $y = \sqrt{\frac{2500 - 4x^2}{4}}$ $= \sqrt{625 - x^2}$		M1	Express y in terms of x	
	$A = 16xy$ $= 16x\sqrt{625 - x^2}$		A1		
Question	Answer	Marks	Partial Marks	Guidance	
11(b)	$\frac{dA}{dx} = 16\sqrt{625 - x^2} + 16x \left[\frac{1}{2}(625 - x^2)^{-\frac{1}{2}}(-2x) \right]$ $= 16\sqrt{625 - x^2} - \frac{16x^2}{\sqrt{625 - x^2}}$ $= \frac{16(625 - x^2) - 16x^2}{\sqrt{625 - x^2}}$ $= \frac{10000 - 32x^2}{\sqrt{625 - x^2}}$	4	M1 A1	Correct usage of product rule $16\sqrt{625 - x^2} - \frac{16x^2}{\sqrt{625 - x^2}}$ seen or better	Students' Performance (P): Average performance, with most candidates scoring 2 out of 4 Common mistakes (CM): - Did not provide exact value of x / did not rationalize / did not simplify - Did not carry forward the minus sign from -2x and ended up with $16\sqrt{625 - x^2} + \frac{16x^2}{\sqrt{625 - x^2}}$

	$\frac{dA}{dx} = 0$ $\frac{10000 - 32x^2}{\sqrt{625 - x^2}} = 0$ $10000 - 32x^2 = 0$ $x^2 = \frac{10000}{32}$ $x = \pm \sqrt{\frac{625}{2}}$ $x = \sqrt{\frac{625}{2}}$ $= \frac{25}{\sqrt{2}}$ $= \frac{25\sqrt{2}}{2}$		M1 ✓	Set $\frac{dA}{dx} = 0$ and solve	
			A1	C.A.O	

Question	Answer	Marks	Partial Marks	Guidance	
12(a)(i)	Since A and E are midpoints of FH and CH respectively, by midpoint theorem , AB is parallel to FG .	1	B1	The phrase ‘midpoint theorem’ must be seen to award B1	Students’ Performance (P): Surprisingly, most candidates did not do this question, possibly due to this question being the Last Topic for AM. Those candidates who did mainly scored 6 out of 9; they mainly did not score in part (a), but were able to explain clearly (though with more steps than expected) for part (b) and part (c), showing that they are well-versed in alternate segment theorem. Candidates are reminded of the need to refer to the marks allocated to know how many steps to write.
12(a)(ii)	$AE = \frac{1}{2}FC$ (midpoint theorem) Since triangles FCH and AEH are similar, $\frac{\text{Area of triangle AEH}}{\text{Area of triangle FCH}} = \left(\frac{1}{2}\right)^2$ $= \frac{1}{4}$	2	B1	Forming correct ratio	
	Hence, area of $FAEC = \frac{3}{4}$ area of triangle FCH Area of $FAEC$: Area of $HFC = 3 : 4$		B1		
Question	Answer	Marks	Partial Marks	Guidance	
12(b)	Angle $BCG = \text{angle } CDB$ (alternate segment theorem)	3	B1		
	Angle $BCG = \text{angle } CBE$ (alternate angles, $AB \parallel FG$)		B1		
	Angle $CDB = \text{angle } CBE$		B1		
12(c)	Angle $BAC = \text{angle } BCG$ (alternate segment theorem)	3	B1		
	Since angle $BAC = \text{angle } BCG$ and angle $BCG = \text{angle } CBE$ (from (a)), angle $BAC = \text{angle } CBE$		B1	Relationship between angle BAC and angle BCE must be explicitly stated; though students need not say ‘from (a)’.	
	Triangle ABC is isosceles with $AC = BC$		B1	‘Isosceles’ must be seen to award B1	
• Minus 1 for whole question if reason not stated or wrong reasons for any steps • Wrong reasons e.g. “alternate-chord theorem”, “tangent-segment theorem”					

Question	Answer	Marks	Partial Marks	Guidance	
13(a)	<u>Commonly seen Method:</u> $\left(\frac{4}{3x}\right)^{\lg 4} = (6x)^{\lg 2}$ $\left(\frac{4}{3x}\right)^{2\lg 2} = (6x)^{\lg 2}$ $\left[\left(\frac{4}{3x}\right)^2\right]^{\lg 2} = (6x)^{\lg 2}$ $\frac{16}{9x^2} = 6x$ $16 = 54x^3$ $x = \frac{2}{3}$	5	M1 M1 M1 M1 A1	Changing lg 4 to 2 lg 2 Making both sides have the same power (lg 2) Equating the bases Isolating x onto 1 side	<u>Students' Performance (P):</u> Despite the power being a logarithm, many candidates were able to recognize that $\lg 4 = 2 \lg 2$ and proceeded from there. With that, a significant number of candidates were able to score full marks (with a small handful of candidates losing marks due to algebraic slips).

Alternative Method:				
$\left(\frac{4}{3x}\right)^{\lg 4} = (6x)^{\lg 2}$ $\lg \left[\left(\frac{4}{3x}\right)^{\lg 4}\right] = \lg \left[(6x)^{\lg 2}\right]$ $\lg 4 \left(\lg \left(\frac{4}{3x}\right)\right) = \lg 2 (\lg 6x)$ $\lg 4 (\lg 4 - \lg 3x) = \lg 2 (\lg 6x)$ $(\lg 4)^2 - \lg 4 (\lg 3x) = \lg 2 (\lg 2 + \lg 3x)$ $(\lg 4)^2 - \lg 4 (\lg 3x) = (\lg 2)^2 + \lg 2 (\lg 3x)$ $(\lg 4)^2 - (\lg 2)^2 = \lg 3x (\lg 4 + \lg 2)$ $(\lg 4 - \lg 2) (\lg 4 + \lg 2) = \lg 3x (\lg 4 + \lg 2)$ $\lg 4 - \lg 2 = \lg 3x$ $\lg 2 = \lg 3x$ $2 = 3x$ $x = \frac{2}{3}$	5	M1	Log both sides	
		M1	Simplify $\lg \left(\frac{4}{3x}\right)$ to $\lg 4 - \lg 3x$	
		M1	Simplify $\lg 6x$ to $\lg 2 + \lg 3x$	
		M1	Arriving at $\lg 4 - \lg 2 = \lg 3x$	
		A1		
As there are a lot of variations with reference to the way to tackle this question,				
General MS for this question:				
M1 – Power law				
M1 – Quotient / product law				
M1 – Application of laws of indices				
M1 – isolating x on one side				
A1				

Question	Answer	Marks	Partial Marks	Guidance	
13(b)	$\log_2 x + \log_4(x+3) = 3$ $\log_2 x + \frac{\log_2(x+3)}{\log_2 4} = 3$	4	M1	Conversion of either logarithm using change of base formula	<u>Students' Performance (P):</u> A significant number of candidates managed to score full marks for this question. <u>Common mistakes (CM):</u> • Forgot to multiply by 2 throughout, causing the loss of A1
	$\log_2 x + \frac{\log_2(x+3)}{2} = 3$		M1 ✓	Converting $\log_2 4$ to 2	
	$2 \log_2 x + \log_2(x+3) = 6$ $\log_2 x^2(x+3) = 6$ $x^2(x+3) = 64$		M1 ✓	Converting $2 \log_2 x + \log_2(x+3)$ to $\log_2 x^2(x+3)$ / correct use of product rule	
	$x^3 + 3x^2 - 64 = 0$		A1	C.A.O.	

Question	Answer	Marks	Partial Marks	Guidance	
14	Coordinates of $B : (2, 4)$	10	B1	Correct coordinates of B	Students' Performance (P): Despite the lack of scaffolding, a significant number of candidates were able to score at least 6 out of 10, demonstrating their ability to differentiate well. However, once candidates start to integrate to find the area of the region, algebraic slips start to appear, causing them to lose their marks accordingly. Common mistakes (CM): Forgot to include 5 when integrating
	$\frac{dy}{dx} = \frac{1}{2}(5x+6)^{-\frac{1}{2}}(5)$ $= \frac{5}{2\sqrt{5x+6}}$		M1	Correct usage of power law	
	At $x = 2$, gradient = $\frac{5}{8}$		M1✓	Correct value of <i>their</i> gradient at $x = 2$	
	Gradient of normal = $-\frac{8}{5}$		M1✓	Correct value of <i>their</i> gradient of normal	
	Equation of normal : $y = -\frac{8}{5}x + 7.2$		M1✓	Correct equation of normal	
	At C , $y = 0$. $x = 4.5$. $C : (4.5, 0)$		M1✓	<i>Their</i> Coordinates of C	
	At A , $y = 0$, $x = -1\frac{1}{5}$		B1	Coordinates of A	
	Area of shaded region = $\int_{-1\frac{1}{5}}^2 \sqrt{5x+6} \, dx + \frac{1}{2} \times 2.5 \times 4$		M1✓	Finding area of shaded region using <i>their</i> coordinates	
	$= \left[\frac{(5x+6)^{\frac{3}{2}}}{\frac{3}{2}} \right]_{-1\frac{1}{5}}^2 + 5$		M1	Correct integration of $\sqrt{5x+6}$	
	$= \frac{2}{15} \left[16^{\frac{3}{2}} - 0 \right] + 5$ $= 13\frac{8}{15}$		A1	C.A.O.	